

Coupon Bight Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

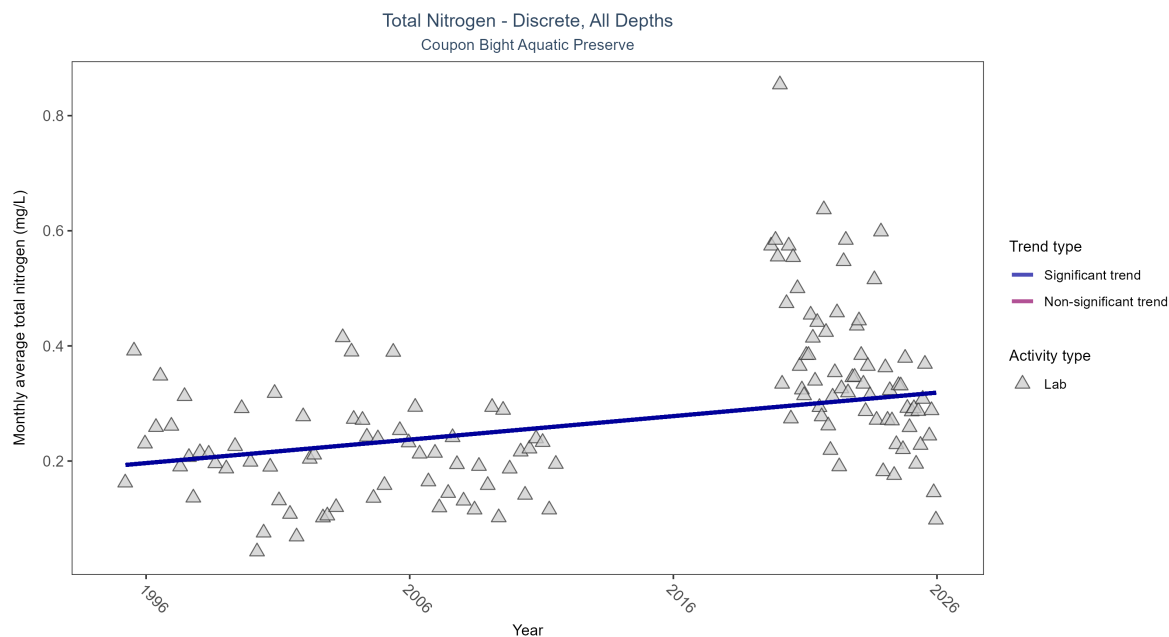


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Monthly average total nitrogen increased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	224	24	1995 - 2025	0.2599	0.2295	0.19234	0.00408	0.0013

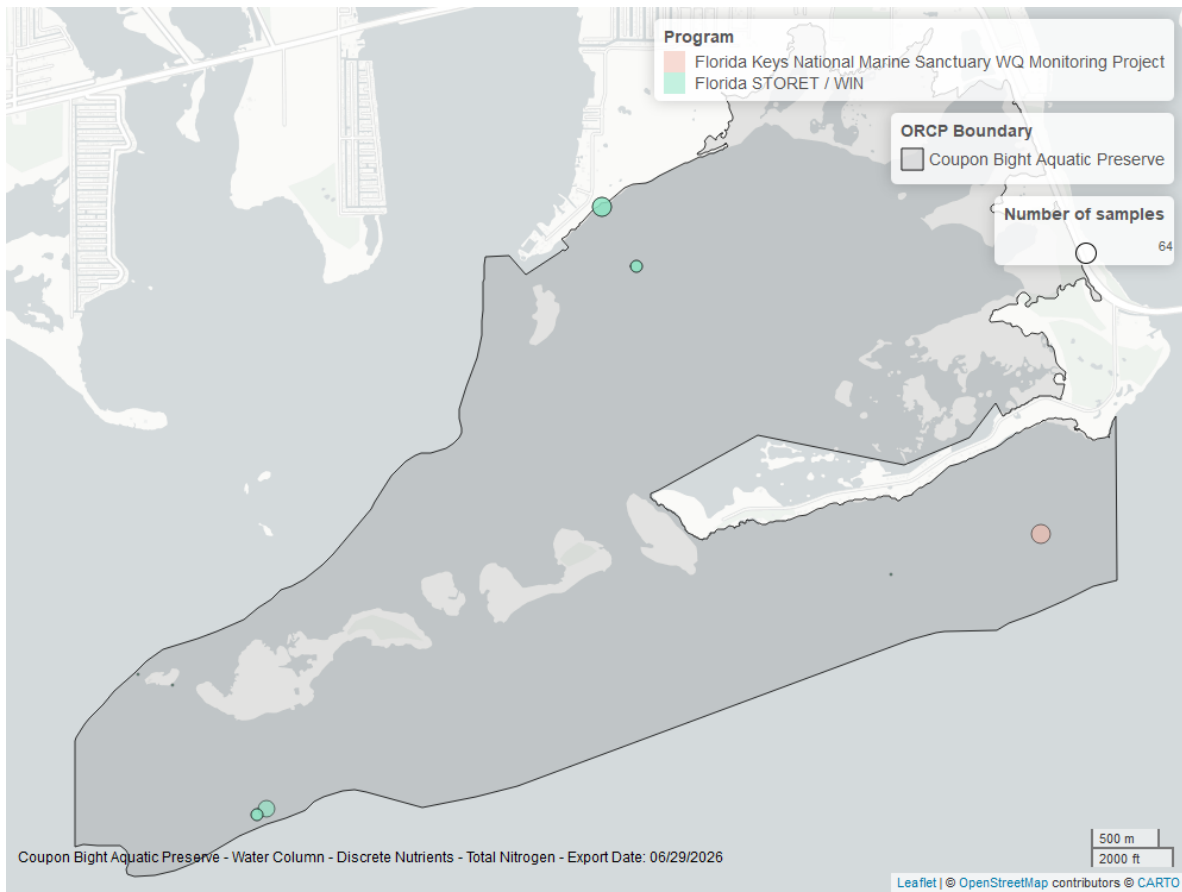


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

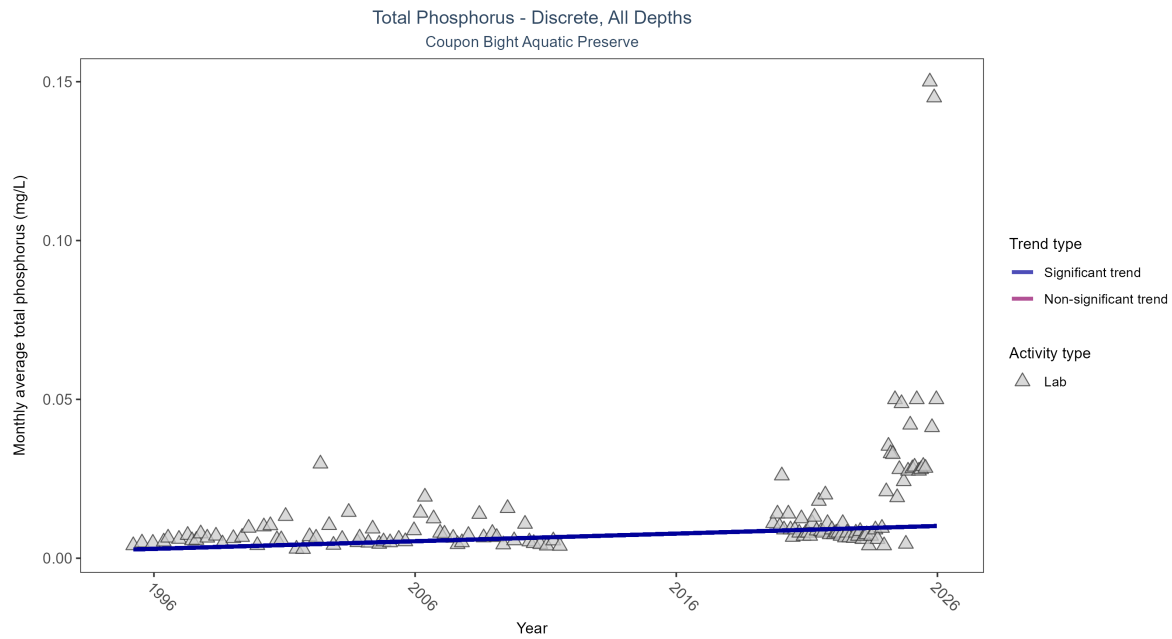


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Monthly average total phosphorus increased by less than 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	225	24	1995 - 2025	0.008	0.41641	0.00271	0.00024	0

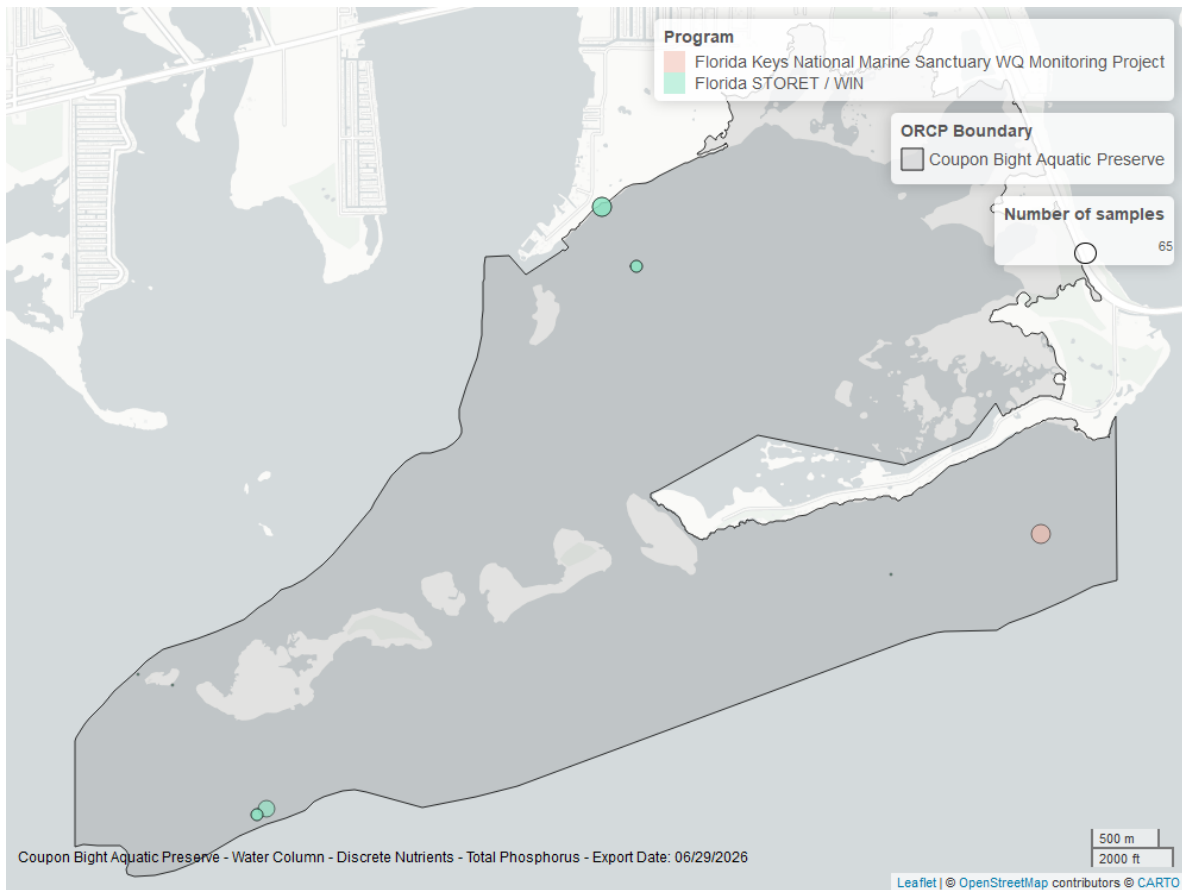


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

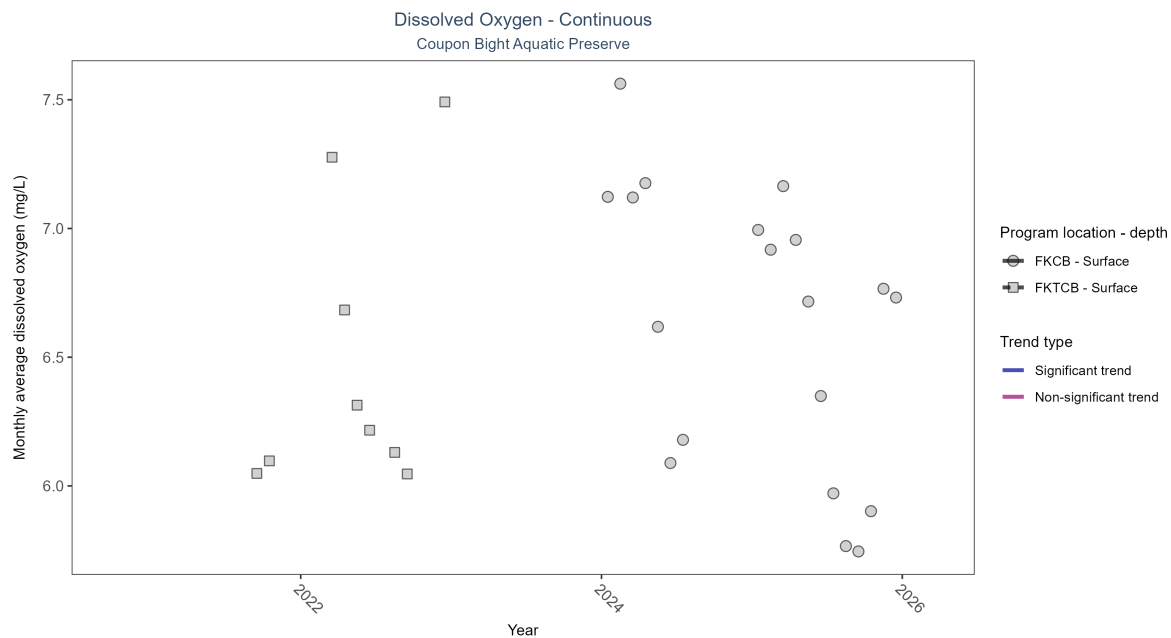


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data	46023	2	2024 - 2025	6.5	-	-	-	-
FKTCB	Insufficient data	13514	2	2021 - 2022	6.4	-	-	-	-

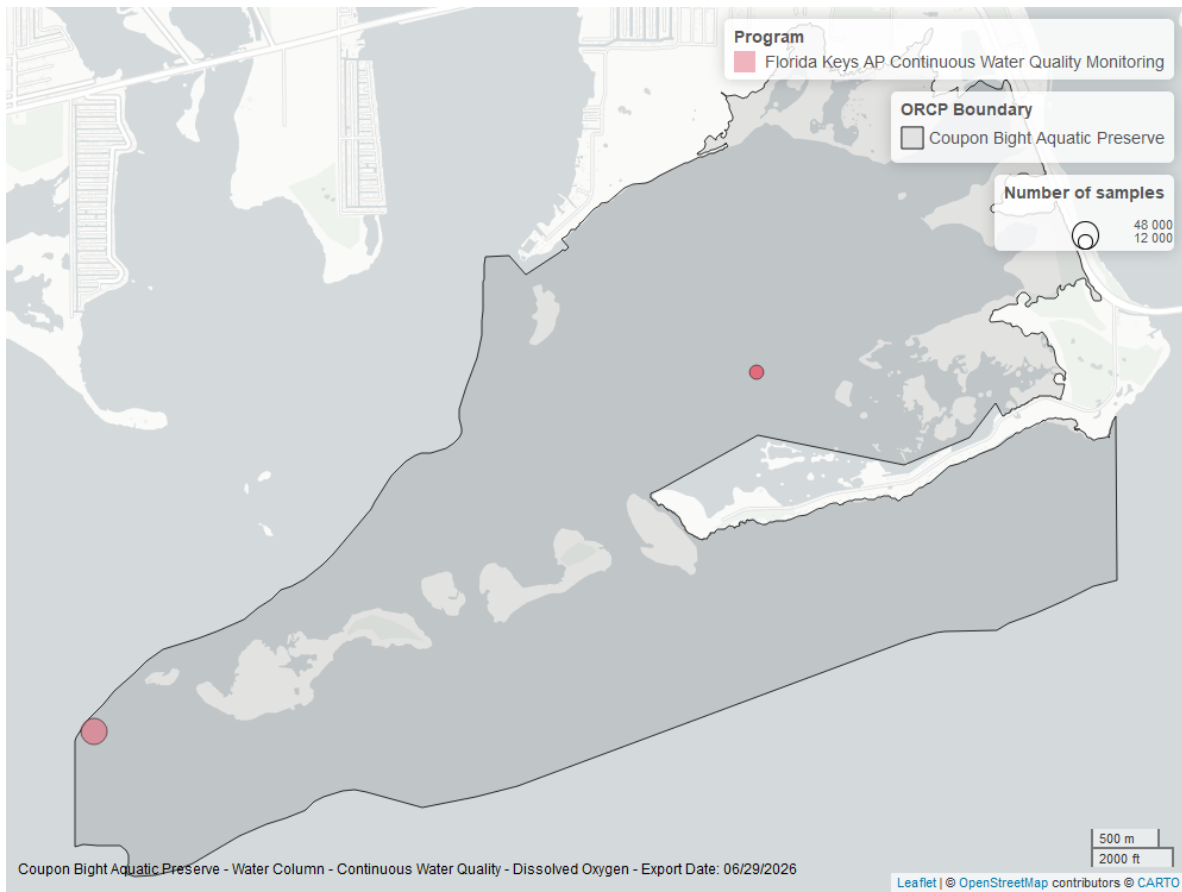


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

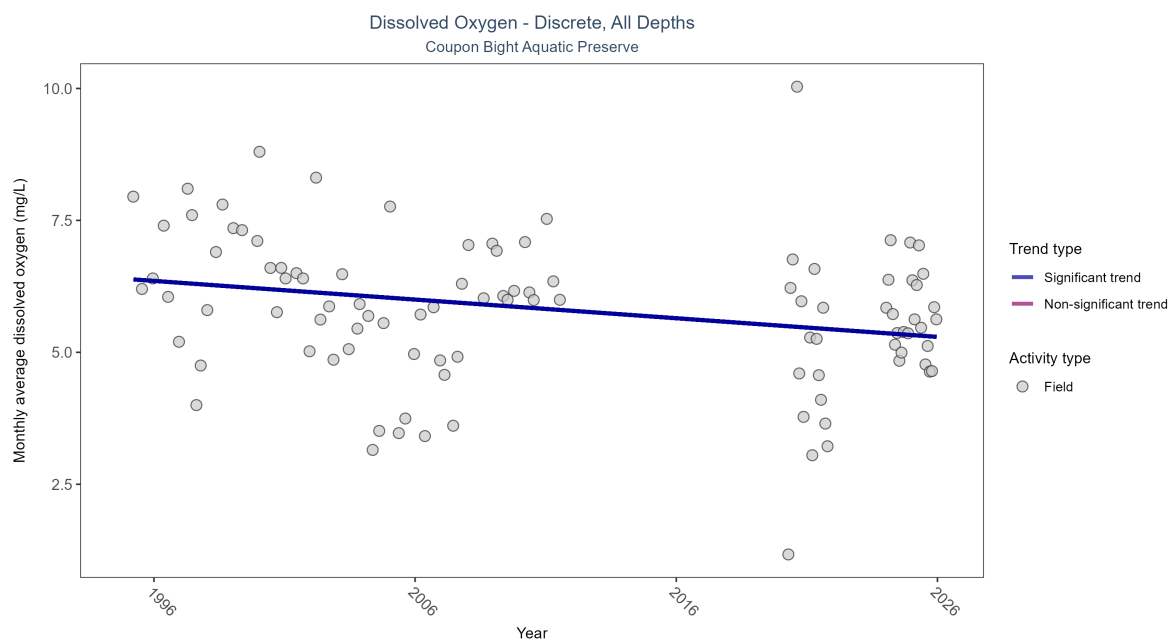


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Monthly average dissolved oxygen decreased by 0.04 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	198	21	1995 - 2025	5.98421	-0.19346	6.38843	-0.03536	0.0179

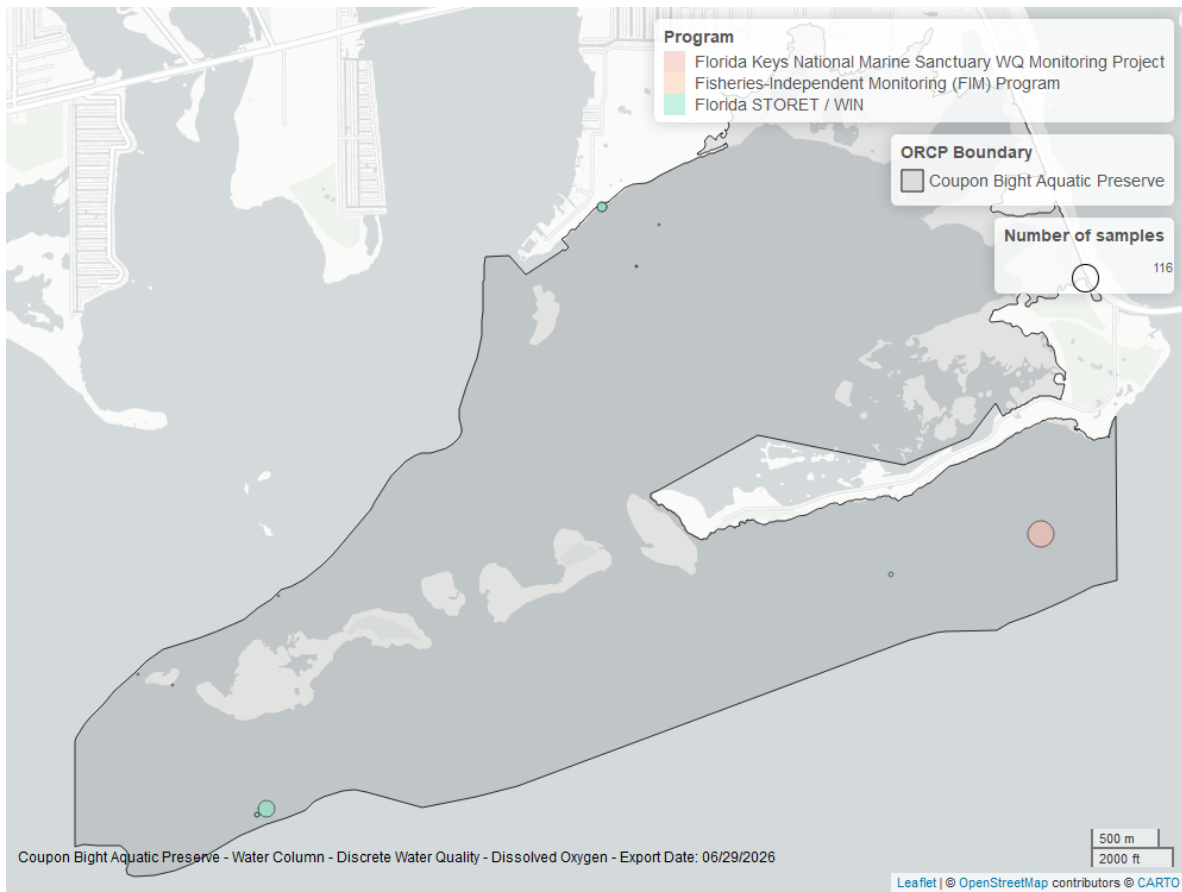


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

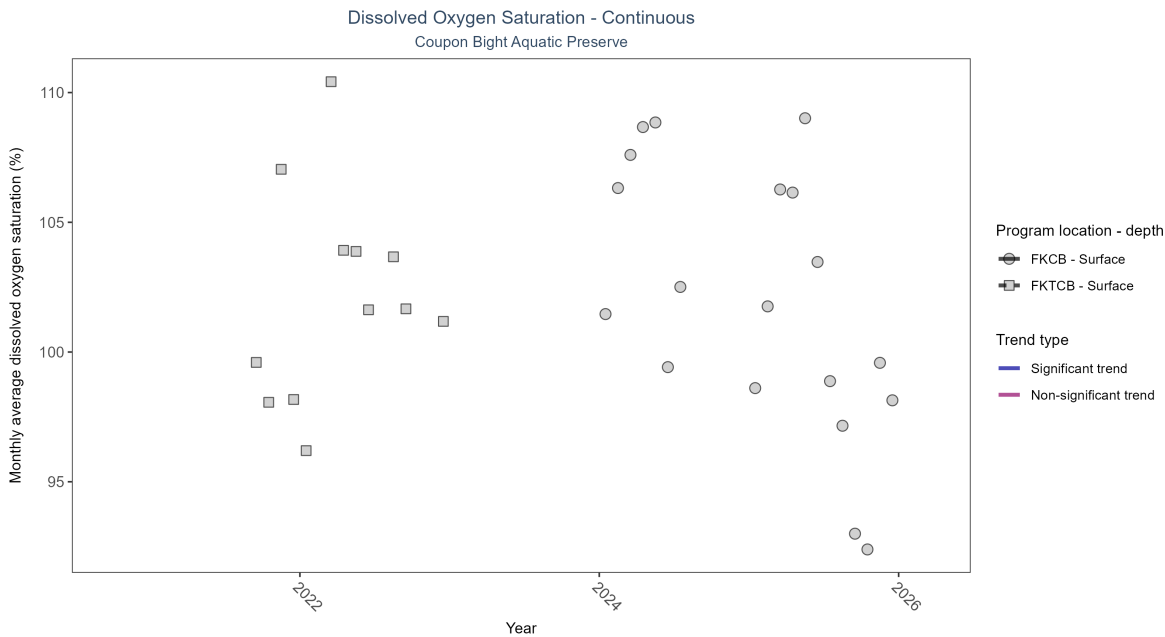


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data	46055	2	2024 - 2025	100.5	-	-	-	-
FKTCB	Insufficient data	17063	2	2021 - 2022	100.2	-	-	-	-

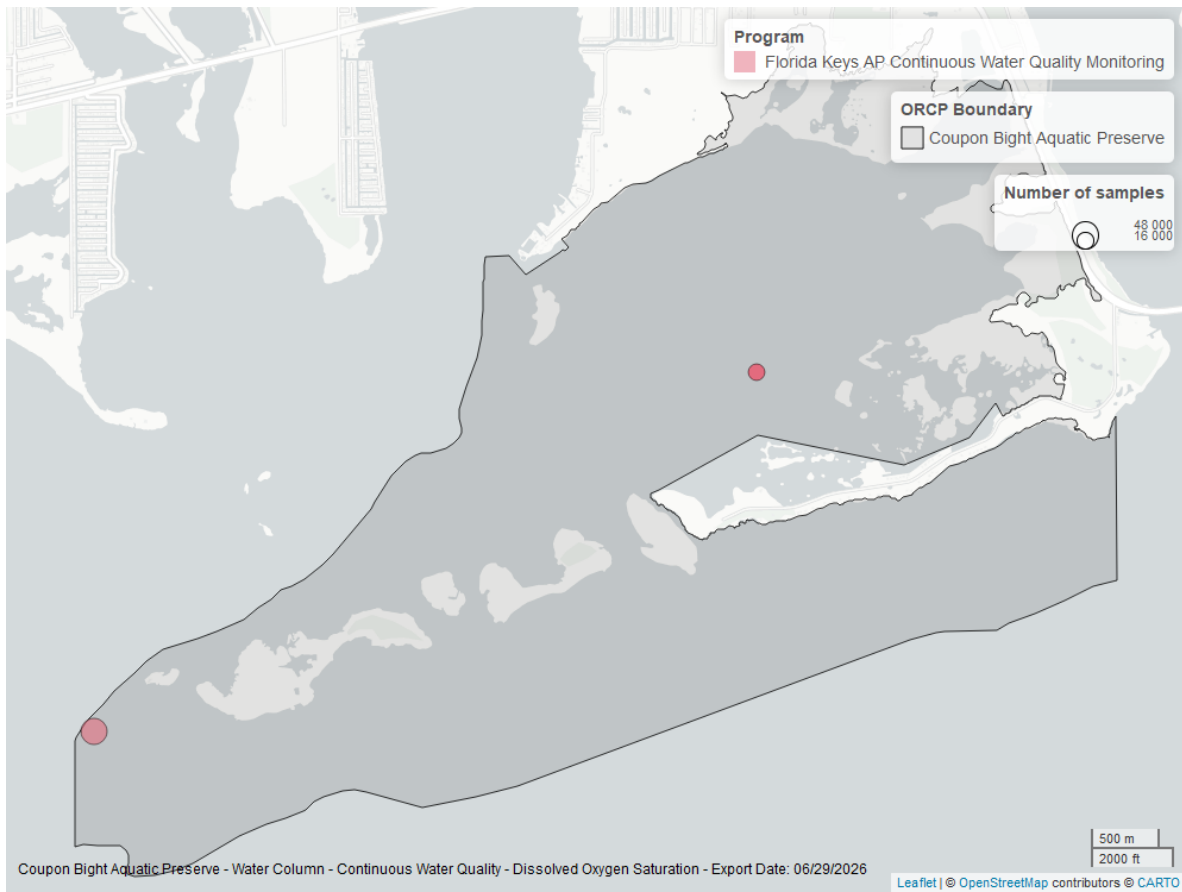


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

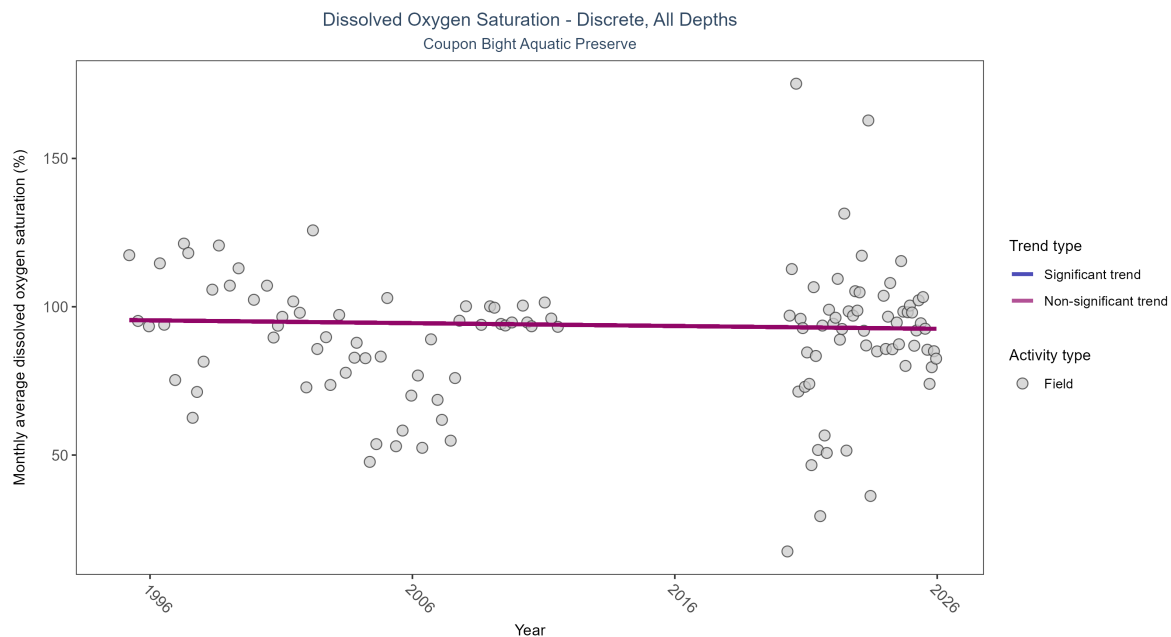


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Dissolved oxygen saturation showed no detectable trend between 1995 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	255	23	1995 - 2025	93.1	-0.05479	95.51341	-0.09535	0.5271

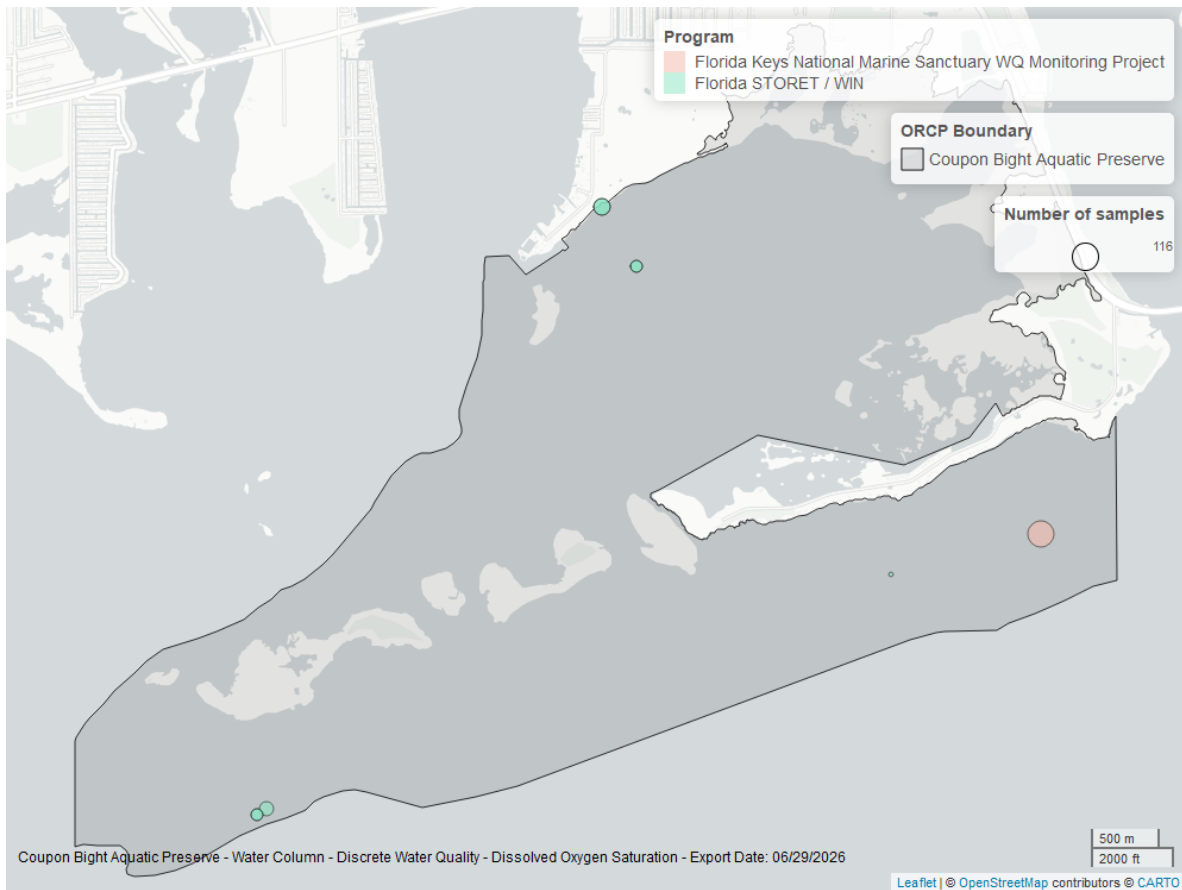


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

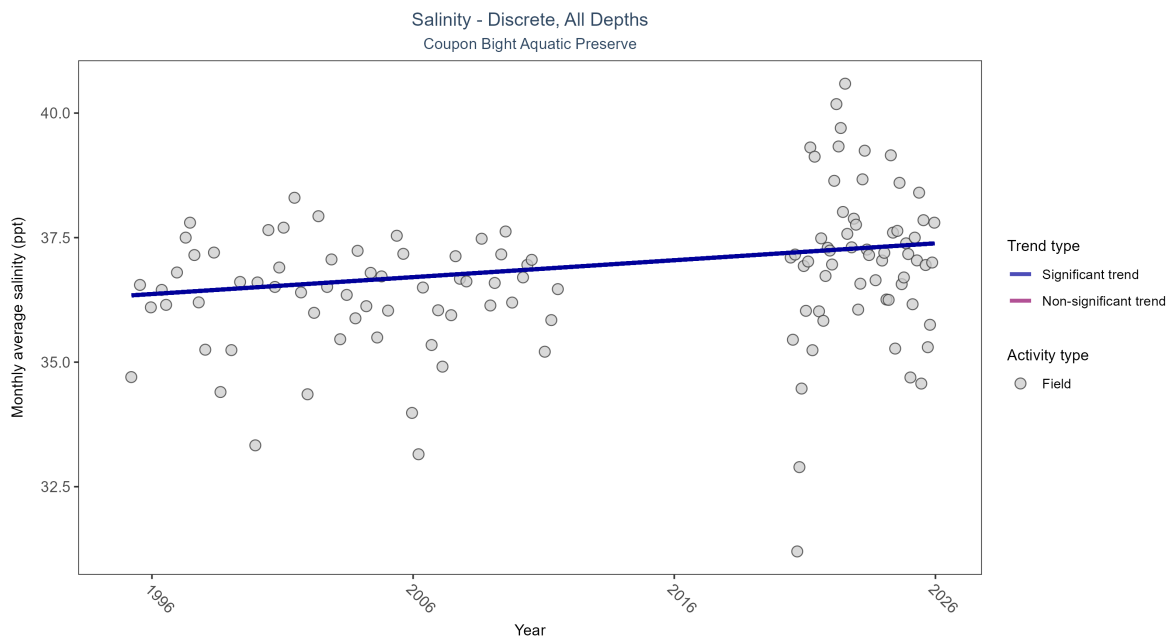


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Monthly average salinity increased by 0.03 ppt per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Increasing trend	259	23	1995 - 2025	36.7537	0.21355	36.33345	0.03393	0.0005

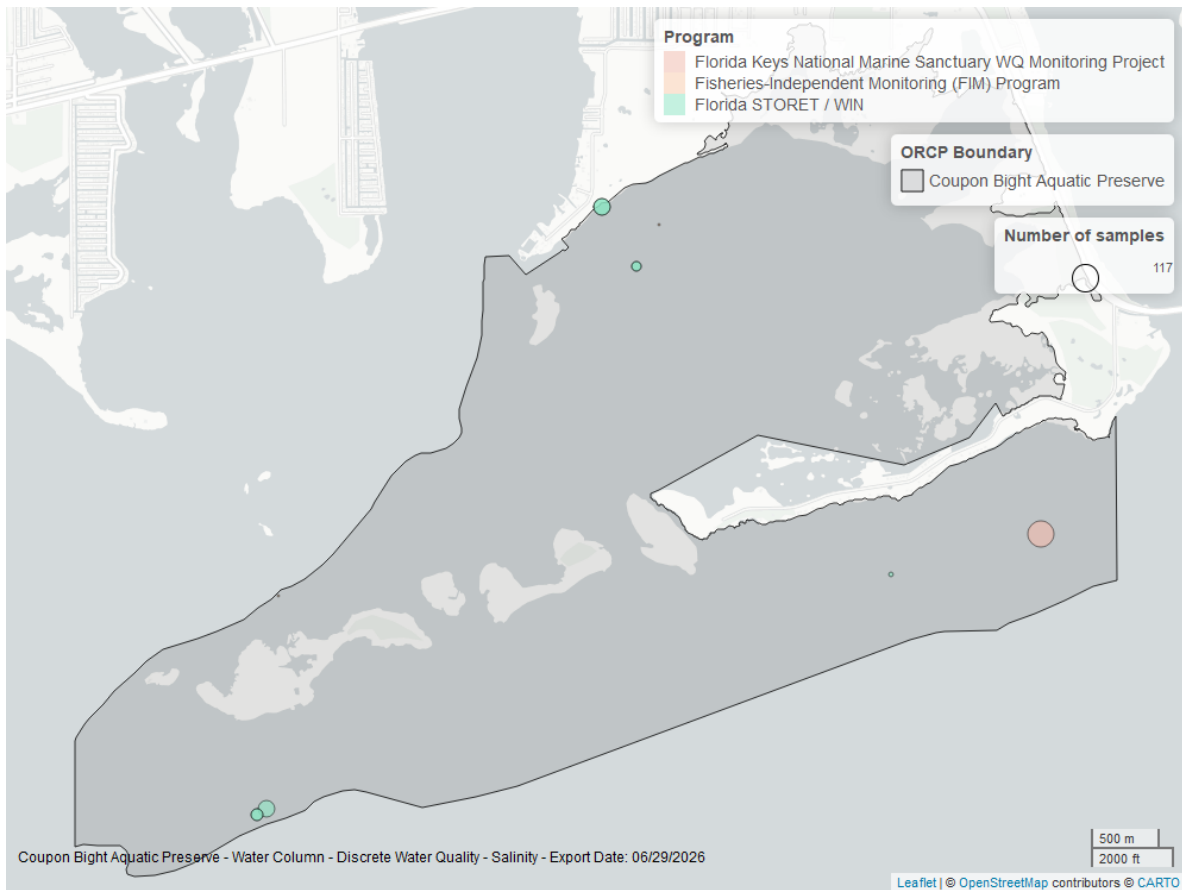


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

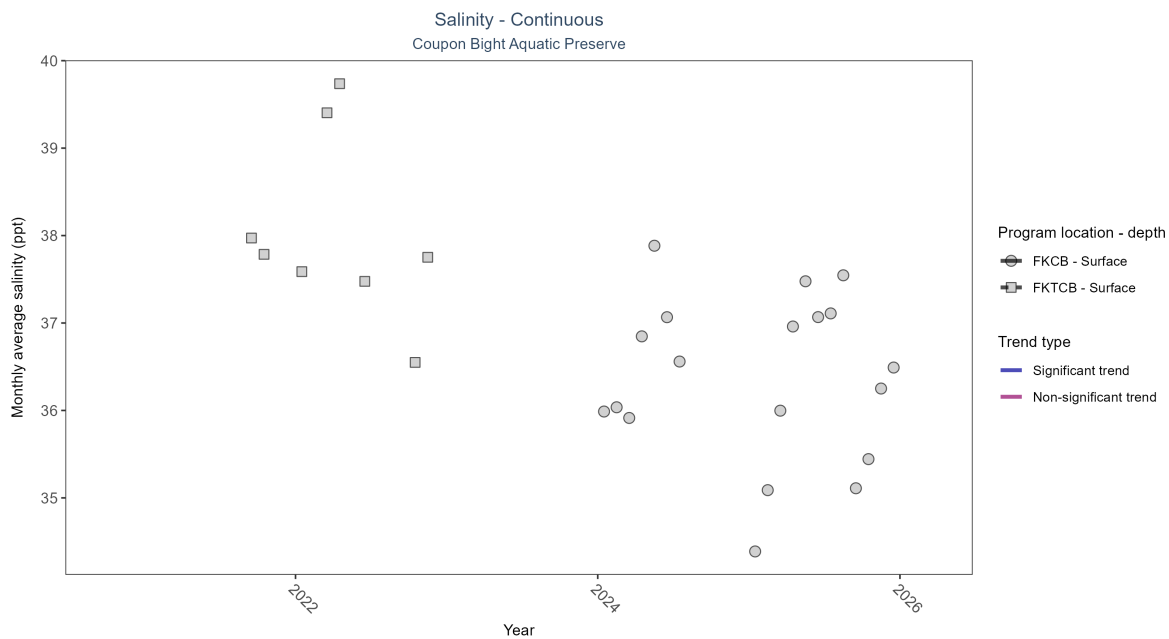


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data	48788	2	2024 - 2025	36.5	-	-	-	-
FKTCB	Insufficient data	13641	2	2021 - 2022	37.9	-	-	-	-

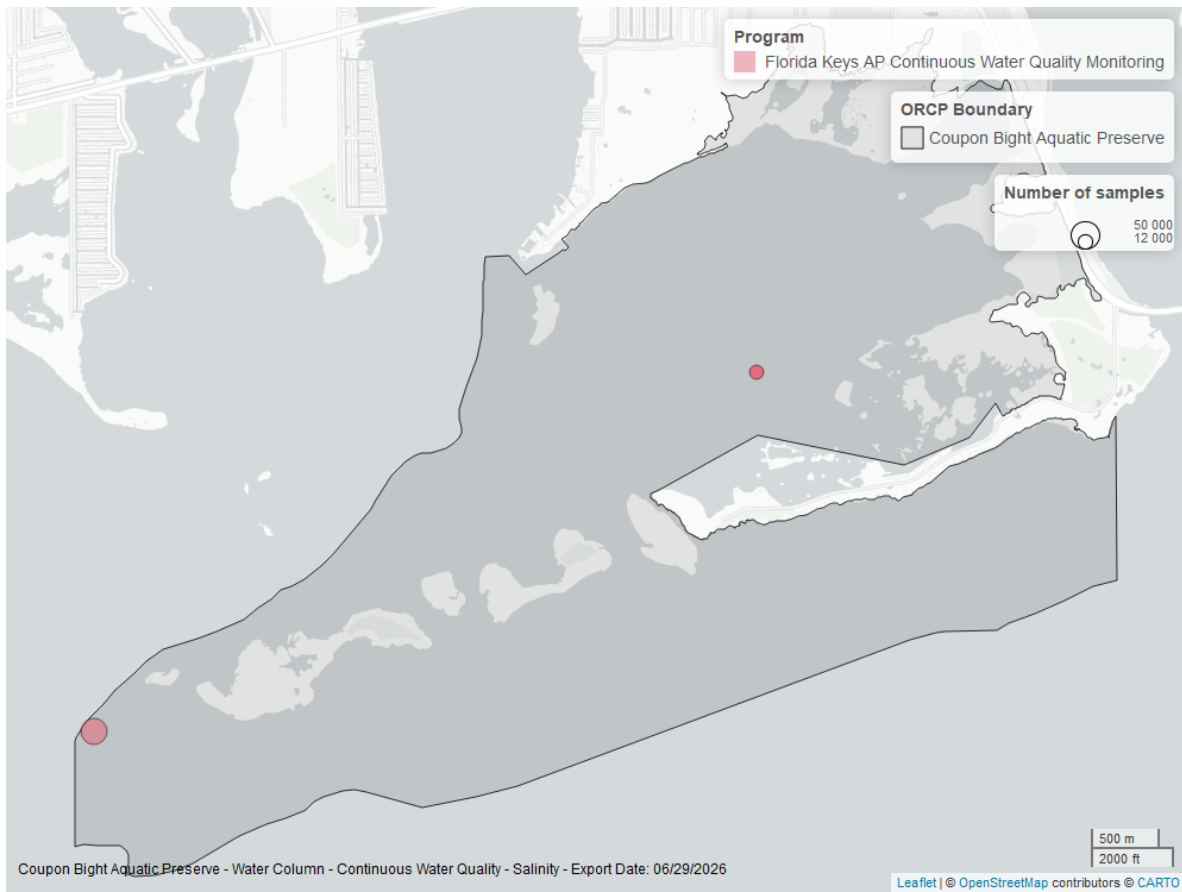


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

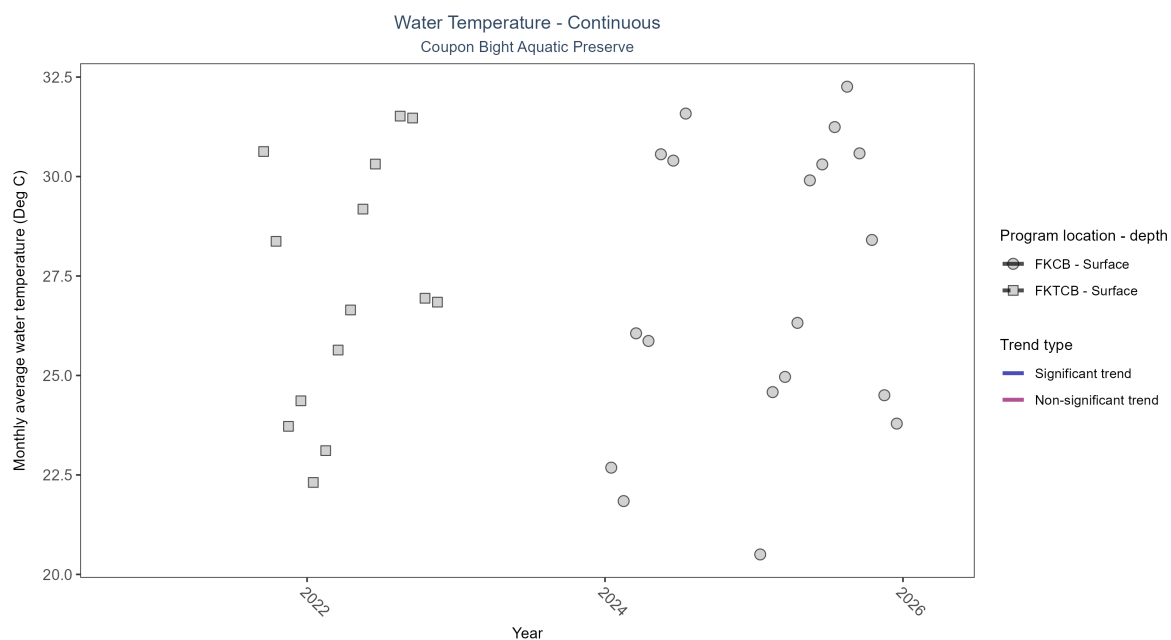


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data	48829	2	2024 - 2025	27.4	-	-	-	-
FKTCB	Insufficient data	30145	2	2021 - 2022	26.7	-	-	-	-

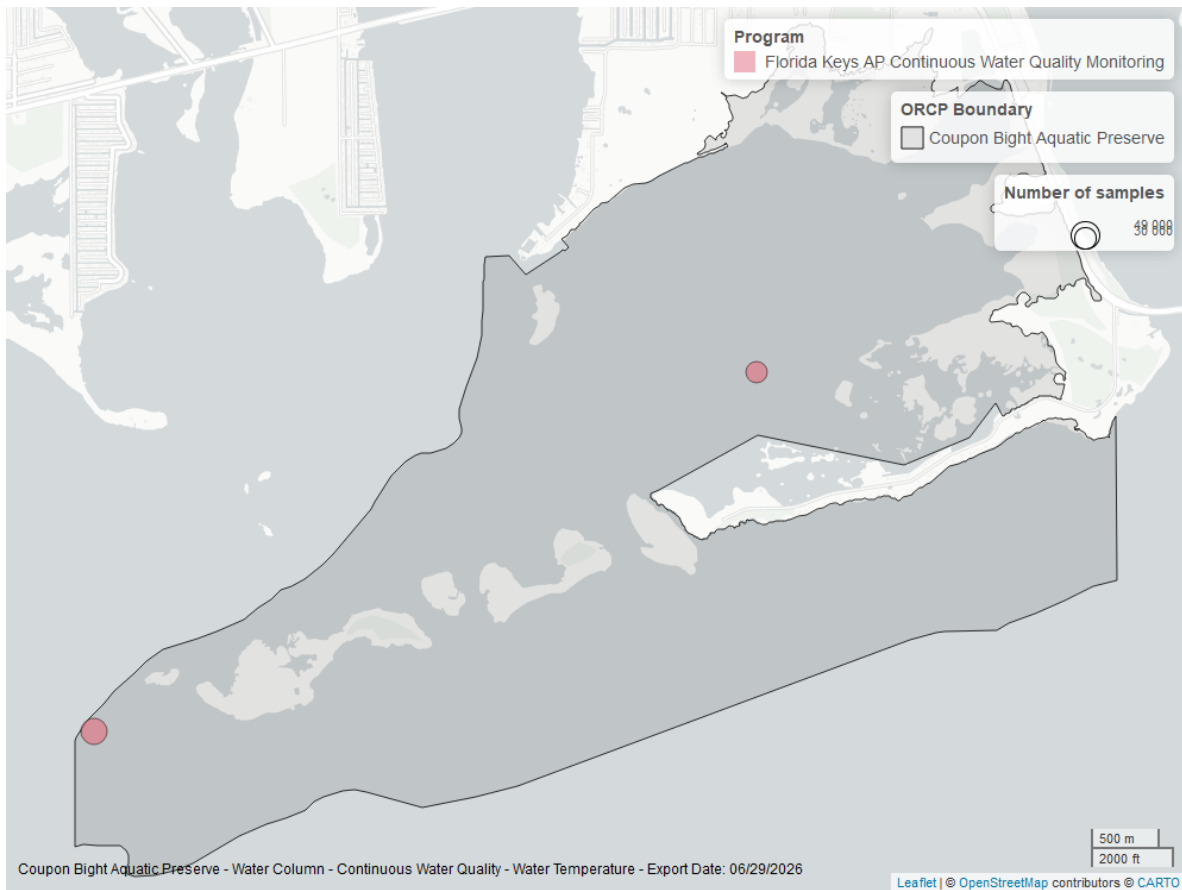


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

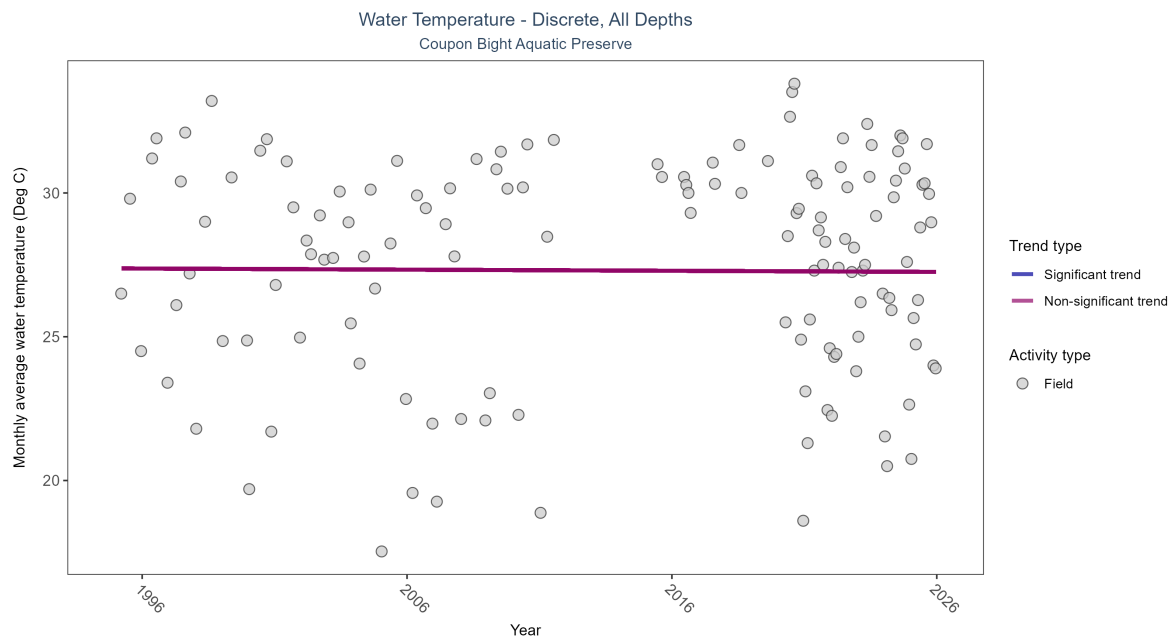


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Water temperature showed no detectable trend between 1995 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	307	28	1995 - 2025	28.88889	-0.01426	27.37571	-0.0039	0.6503

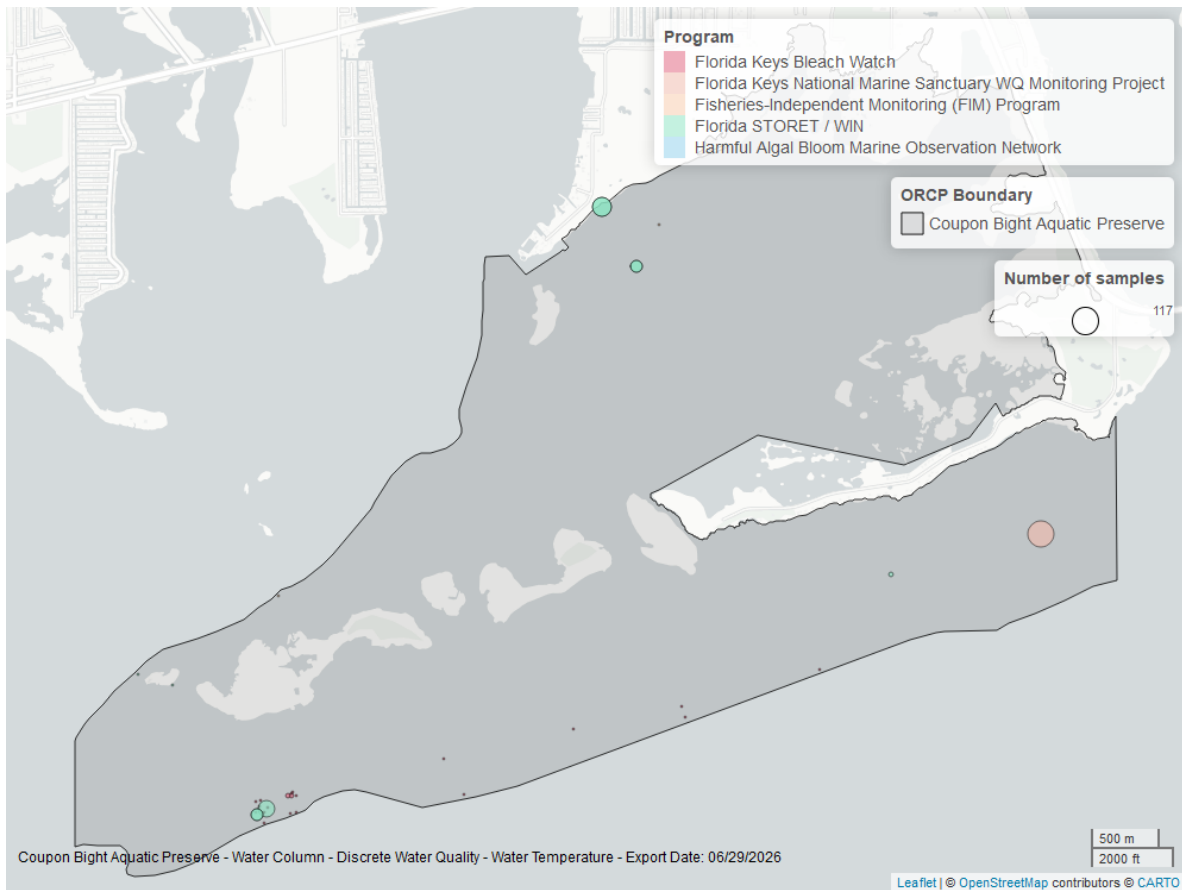


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

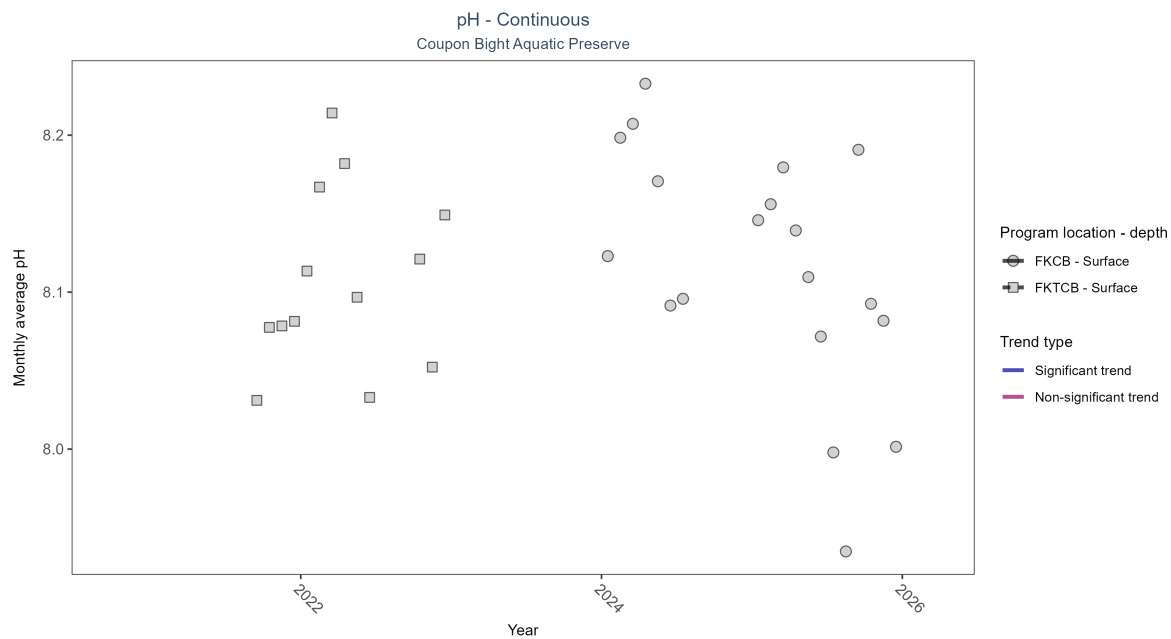


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data	48063	2	2024 - 2025	8.1	-	-	-	-
FKTCB	Insufficient data	22914	2	2021 - 2022	8.1	-	-	-	-

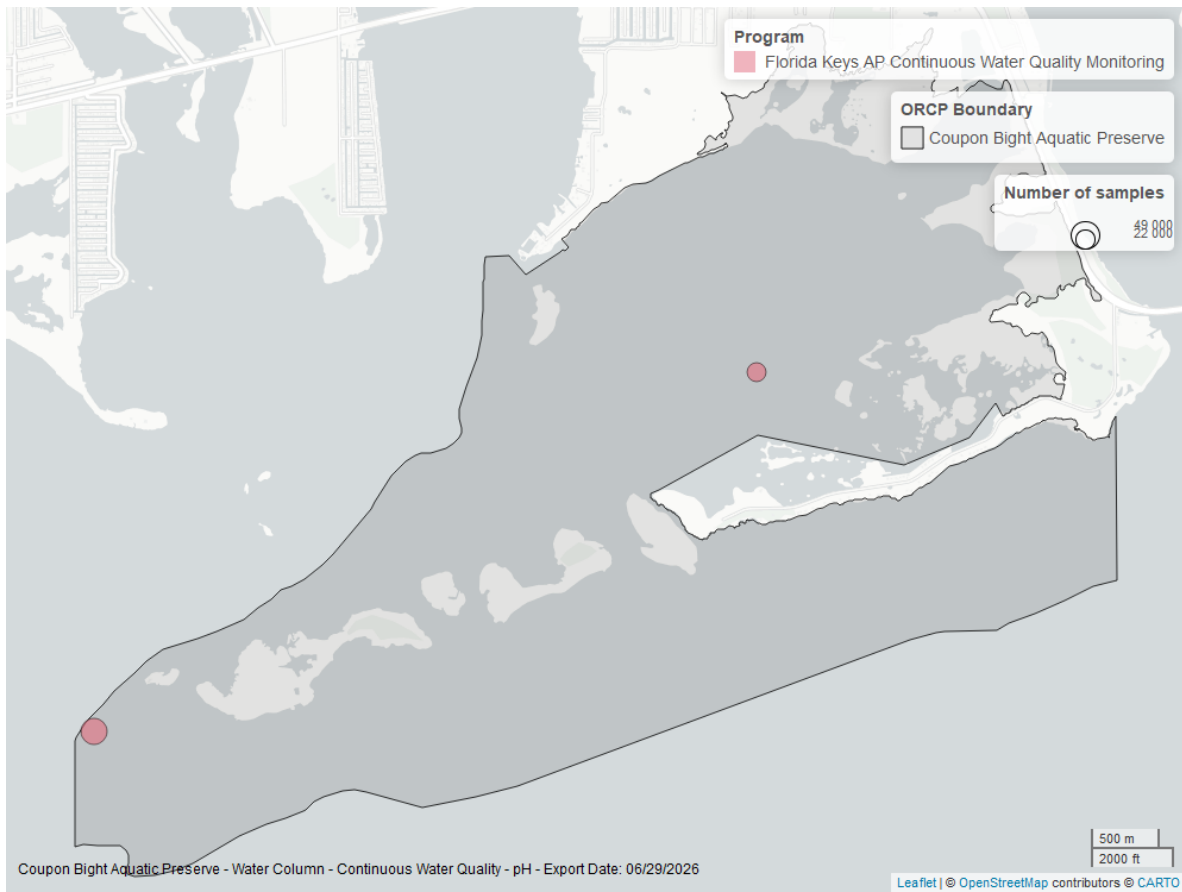


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

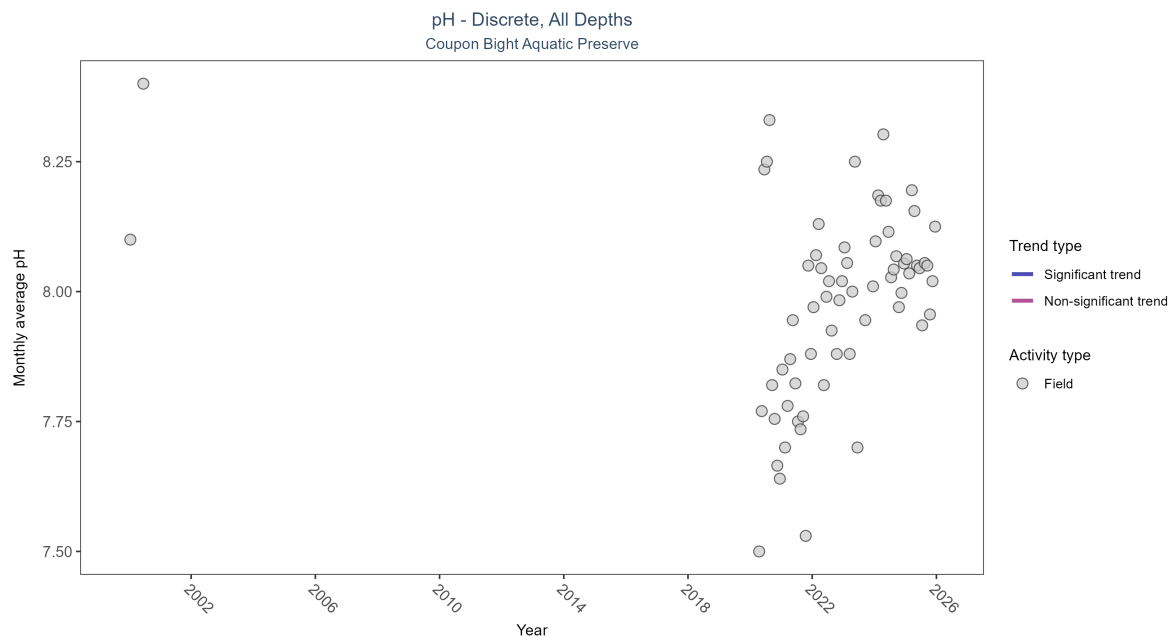


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: There was insufficient data to fit a model for pH.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Insufficient data	158	7	2000 - 2025	8.03	-	-	-	-

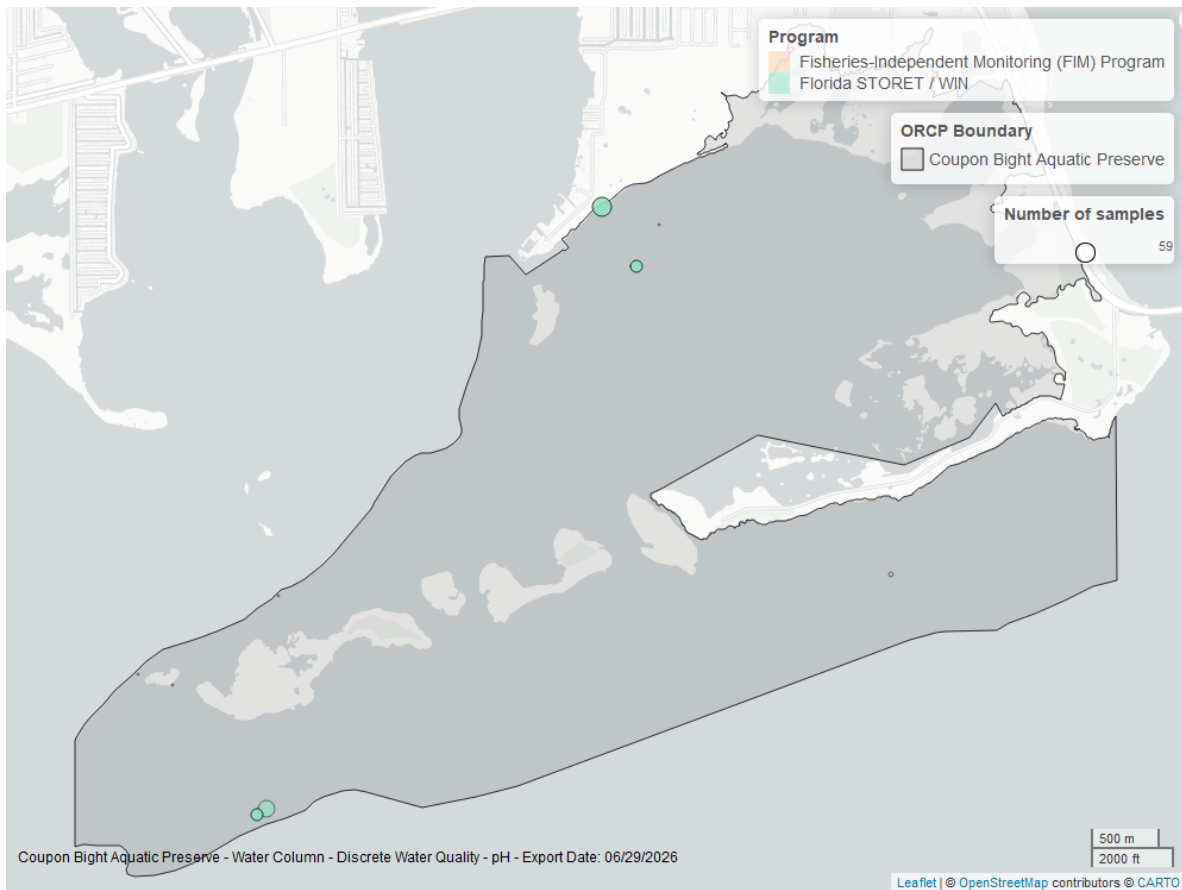


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

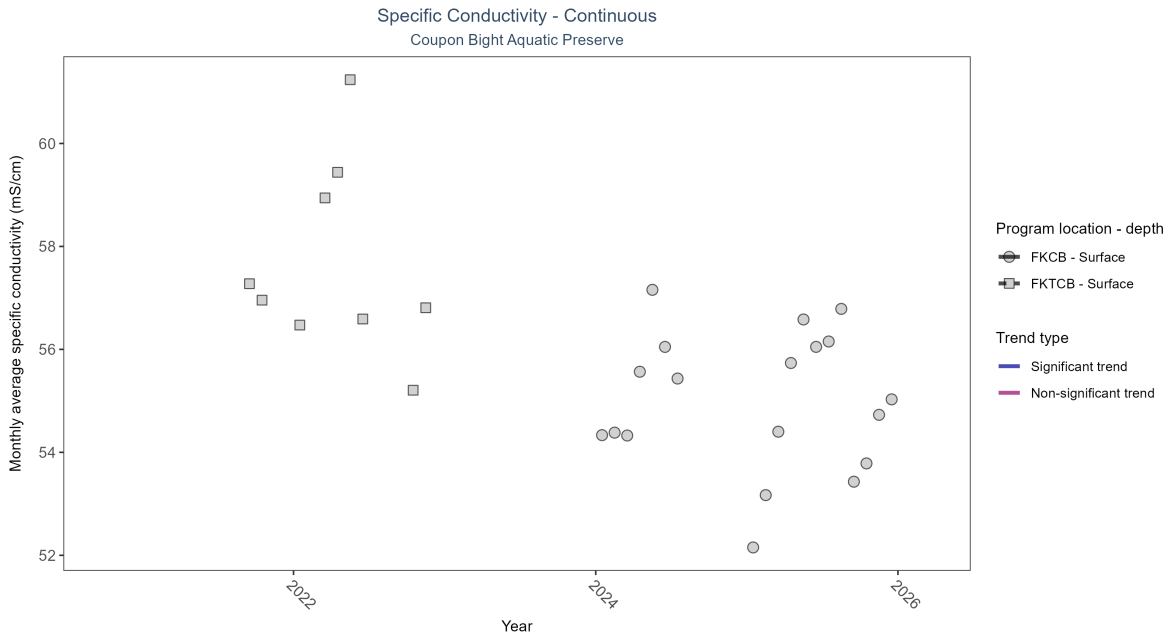


Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data	48788	2	2024 - 2025	55.09	-	-	-	-
FKTCB	Insufficient data	13641	2	2021 - 2022	57.01	-	-	-	-

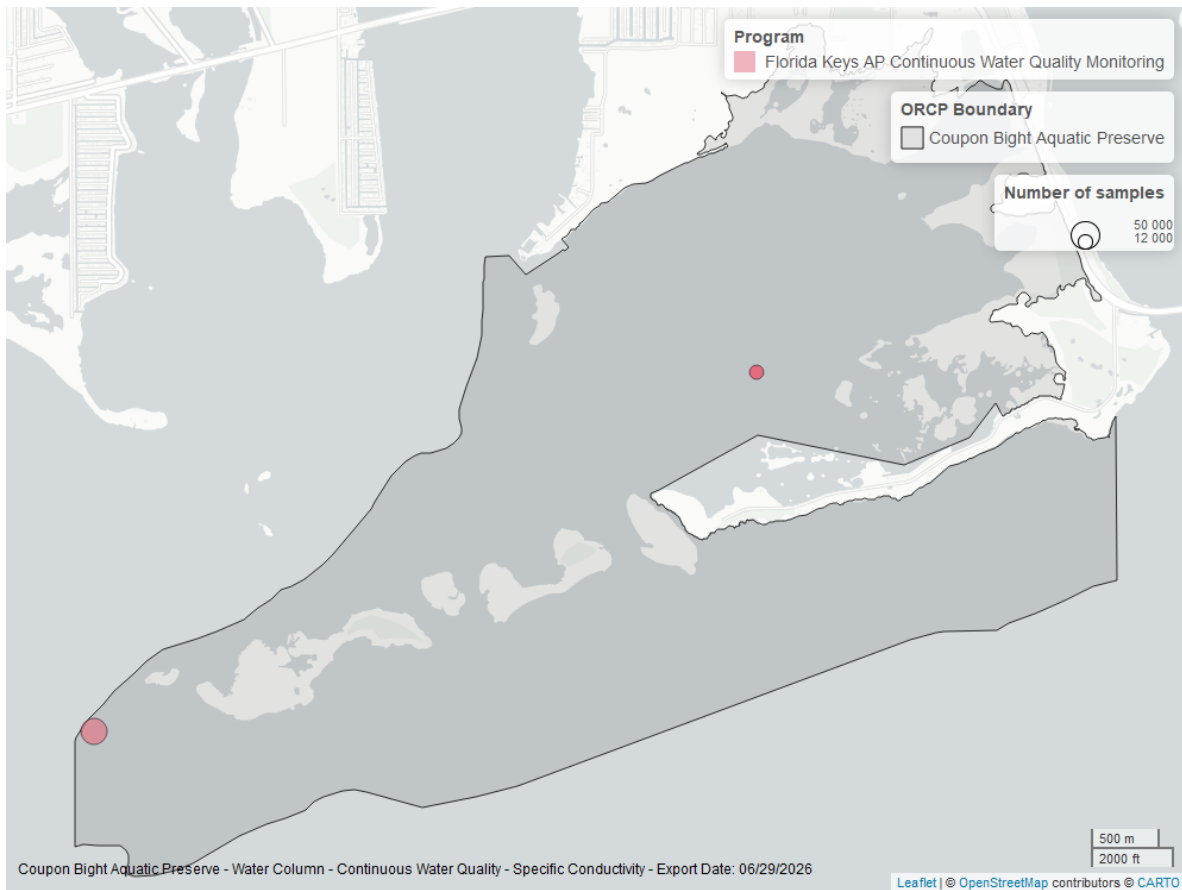


Figure 26: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

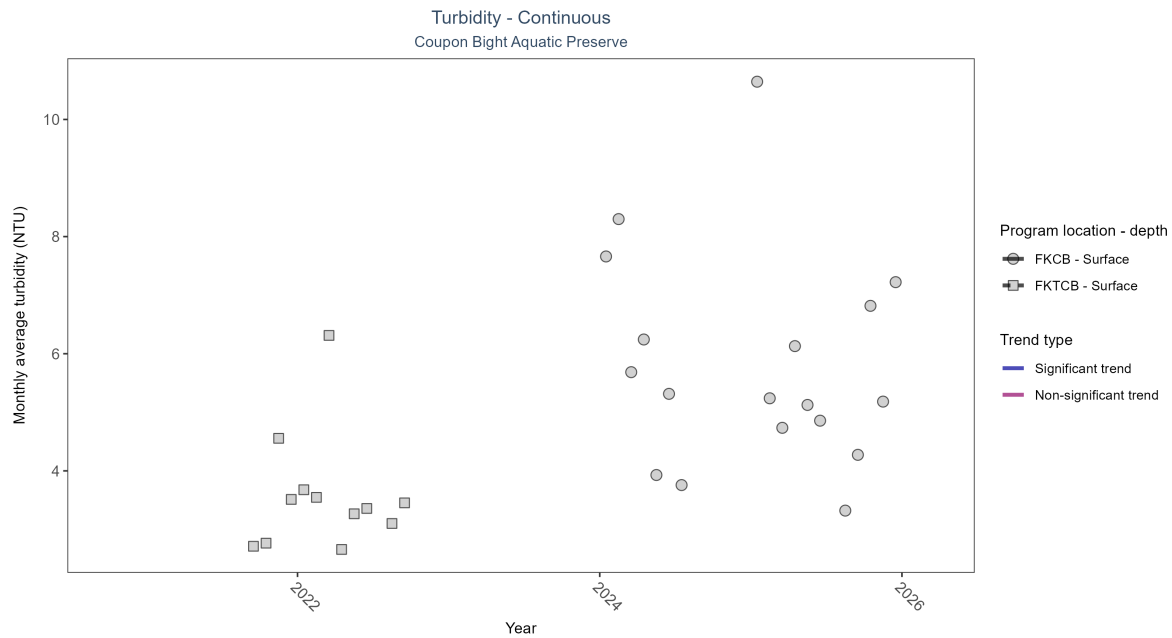


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: There was insufficient data to fit a model for two locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
FKCB	Insufficient data	37724	2	2024 - 2025	4	-	-	-	-
FKTCB	Insufficient data	19289	2	2021 - 2022	3	-	-	-	-

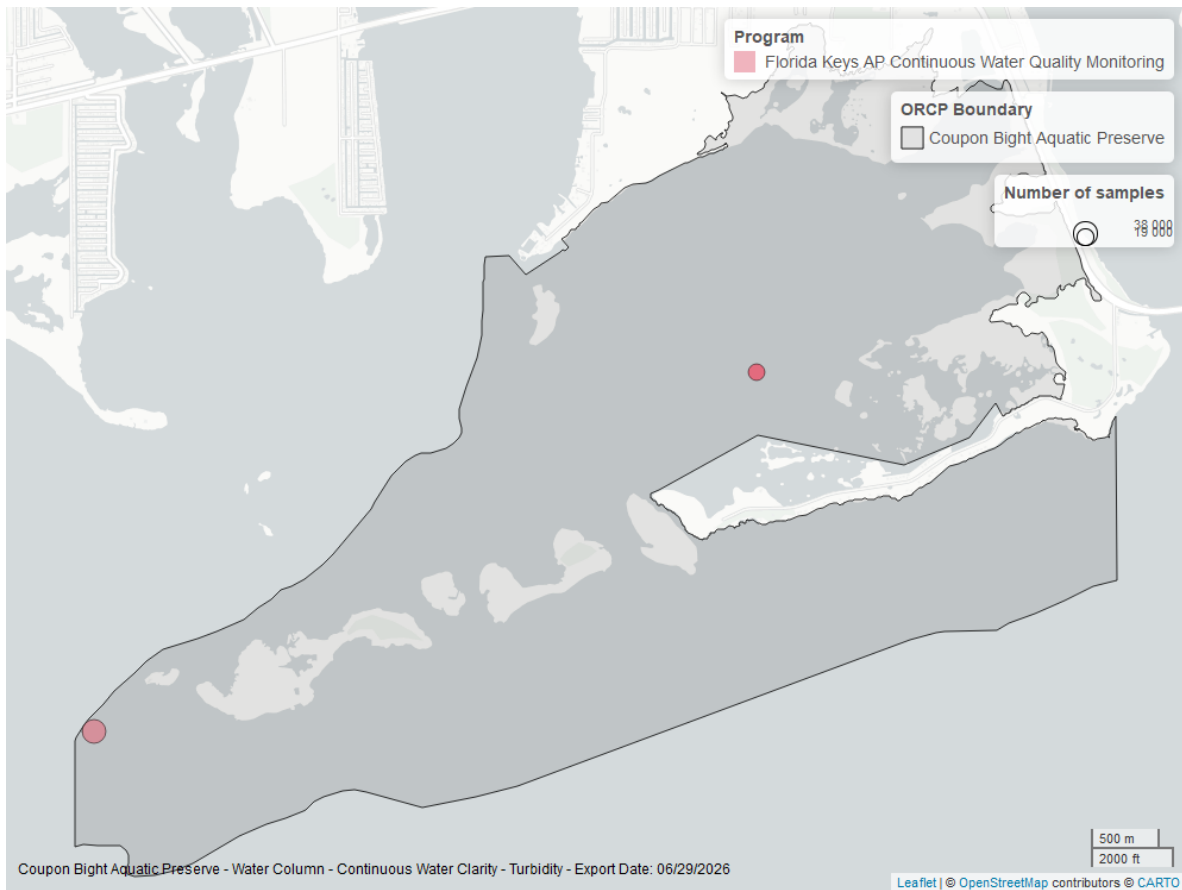


Figure 28: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

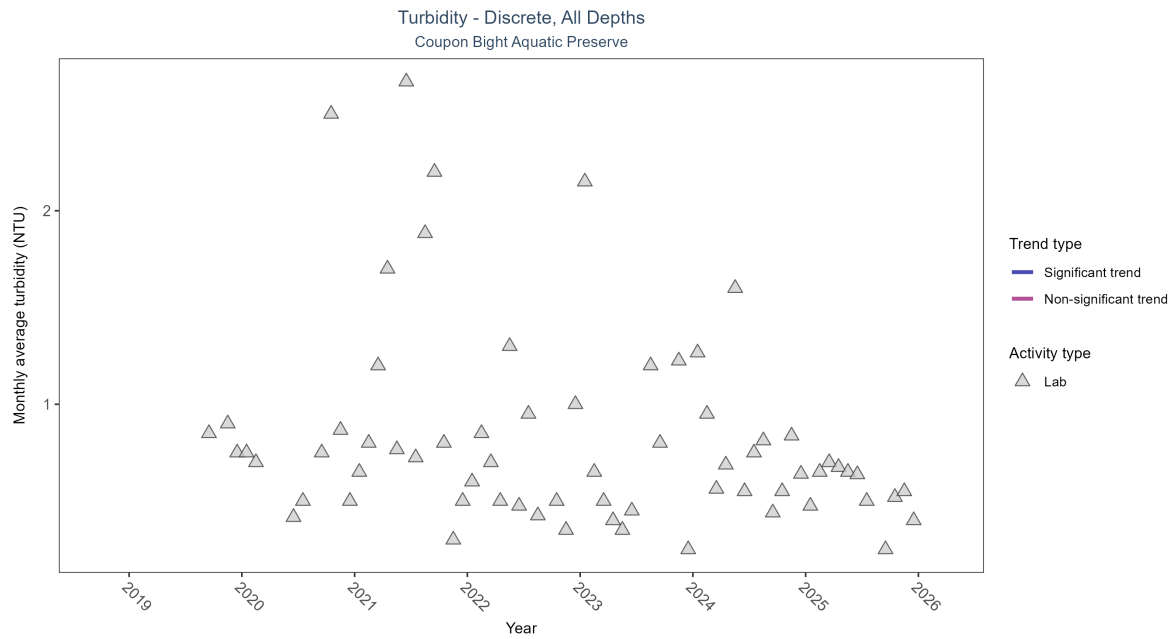


Figure 29: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: There was insufficient data to fit a model for turbidity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	156	7	2019 - 2025	0.6	-	-	-	-

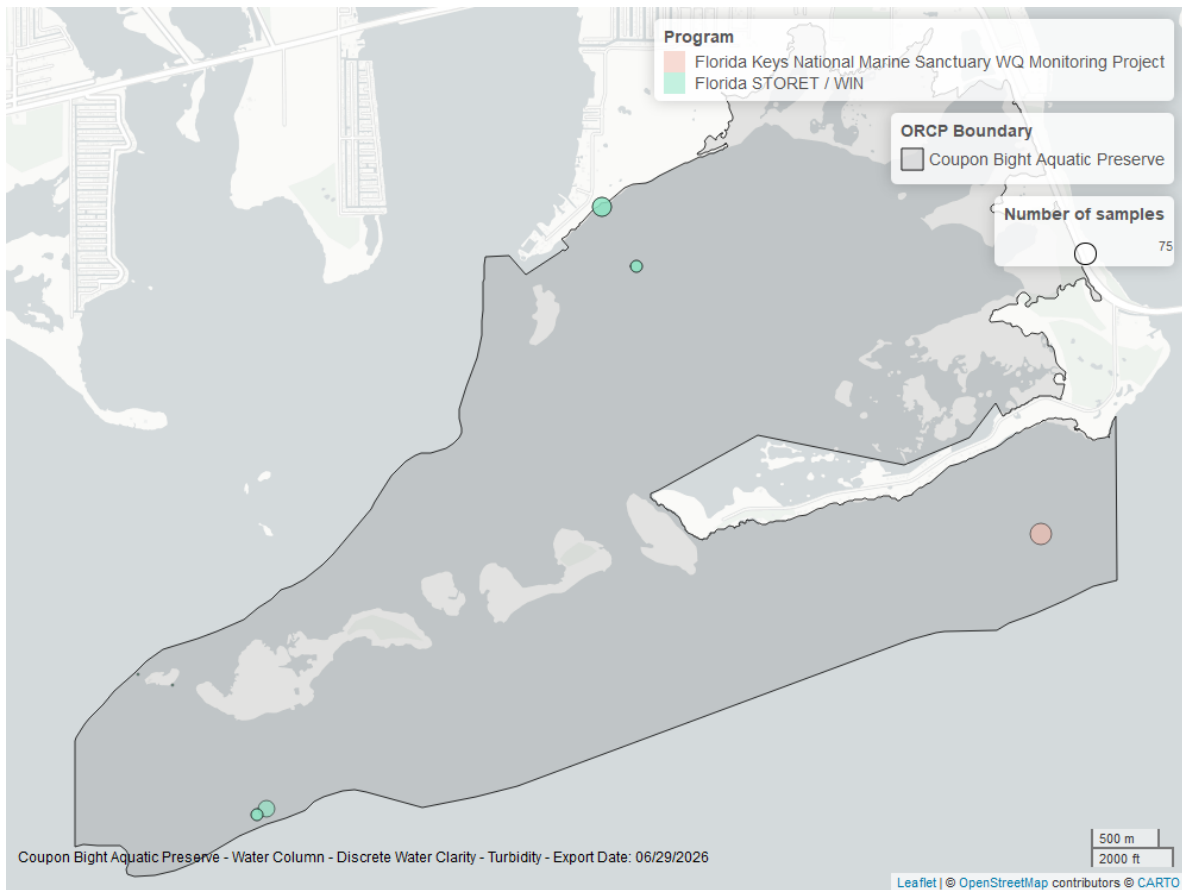


Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

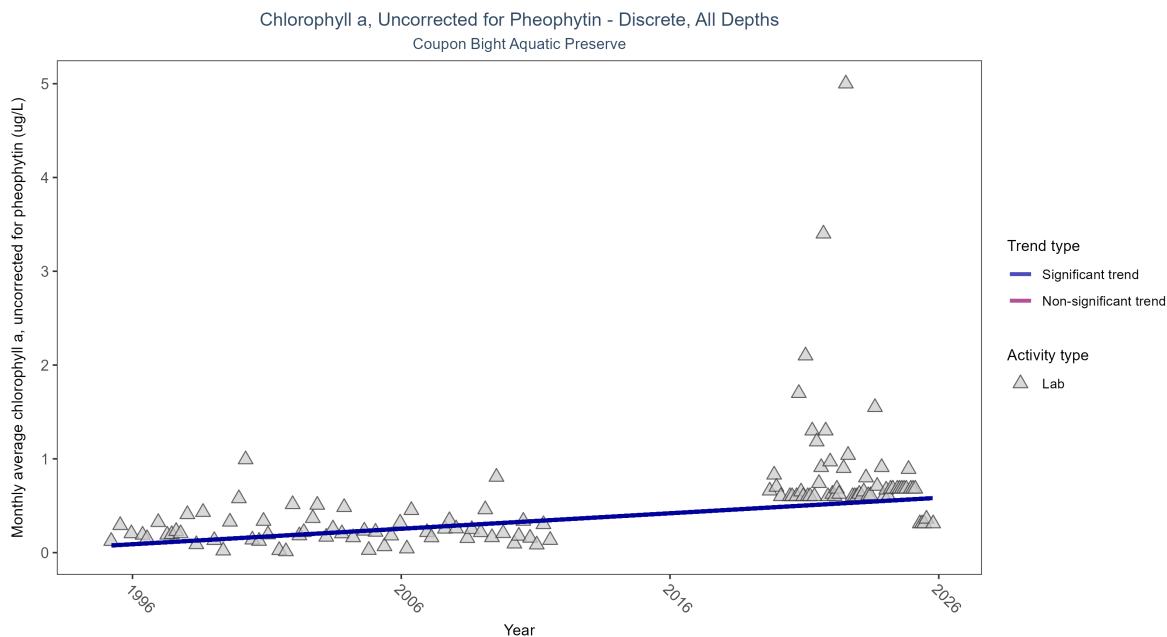


Figure 31: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 16: Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.02 ug/L per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	173	24	1995 - 2025	0.6	0.43574	0.07295	0.01651	0

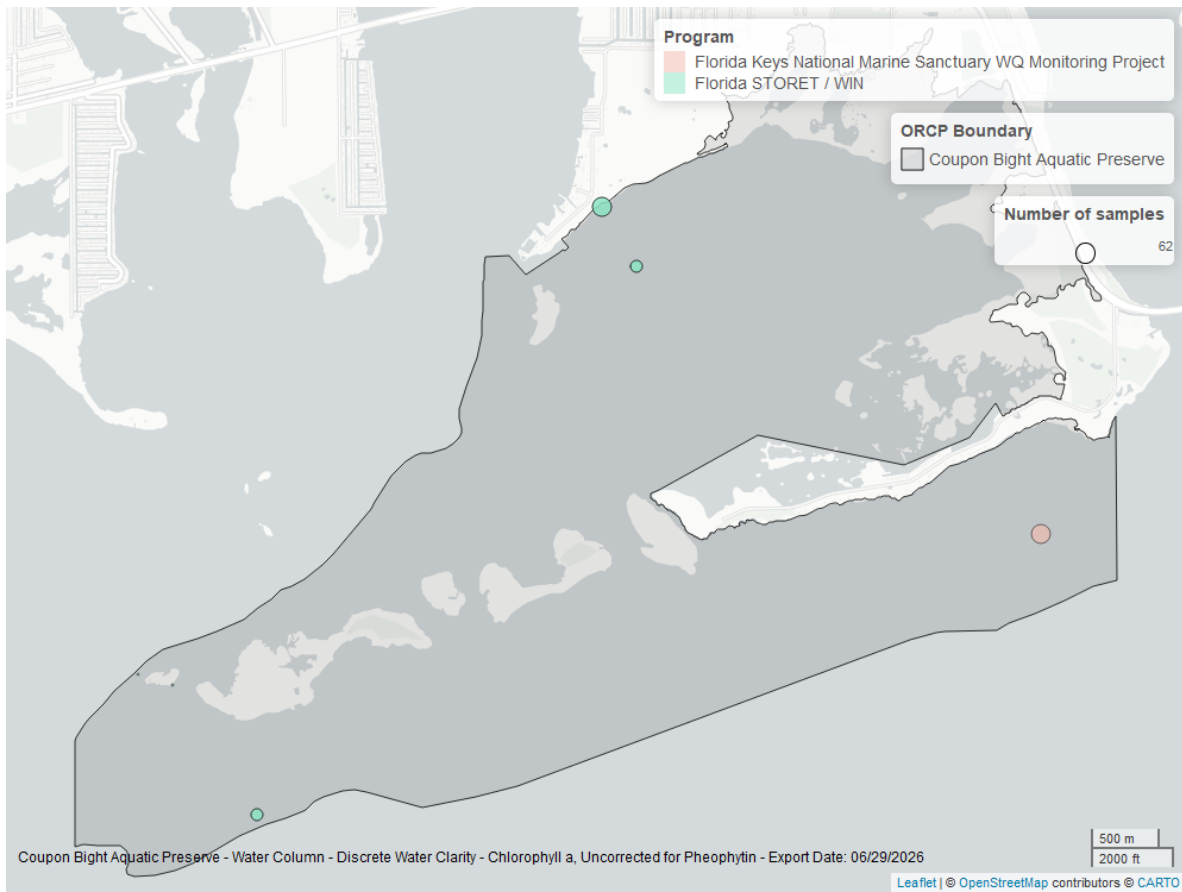


Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

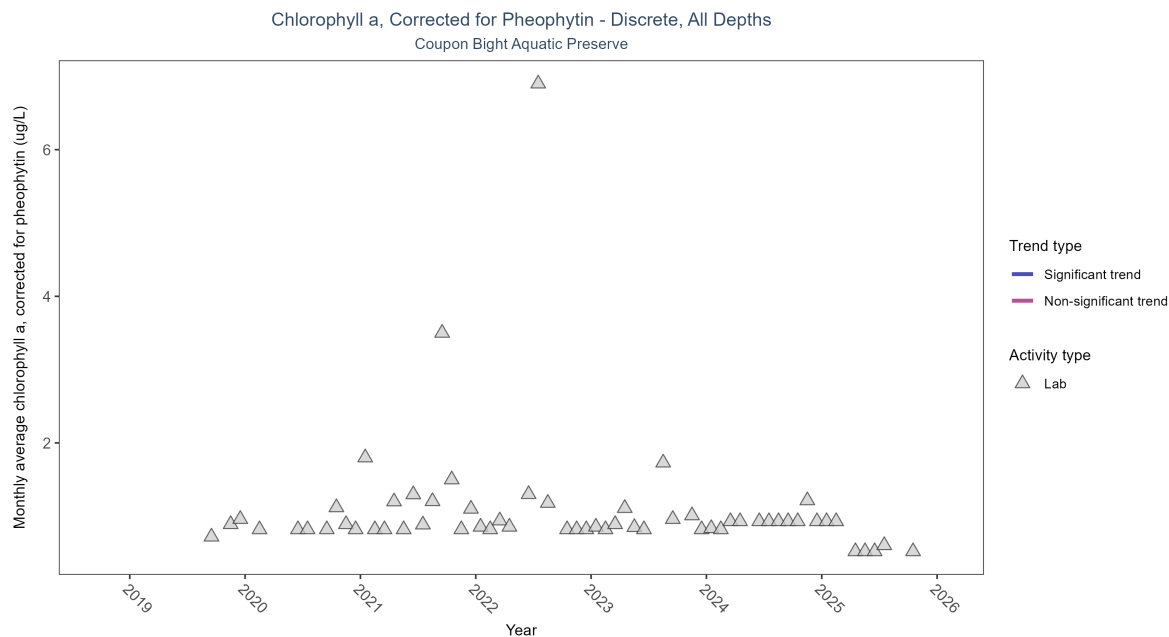


Figure 33: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: There was insufficient data to fit a model for chlorophyll a, corrected for pheophytin.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	113	7	2019 - 2025	0.89	-	-	-	-

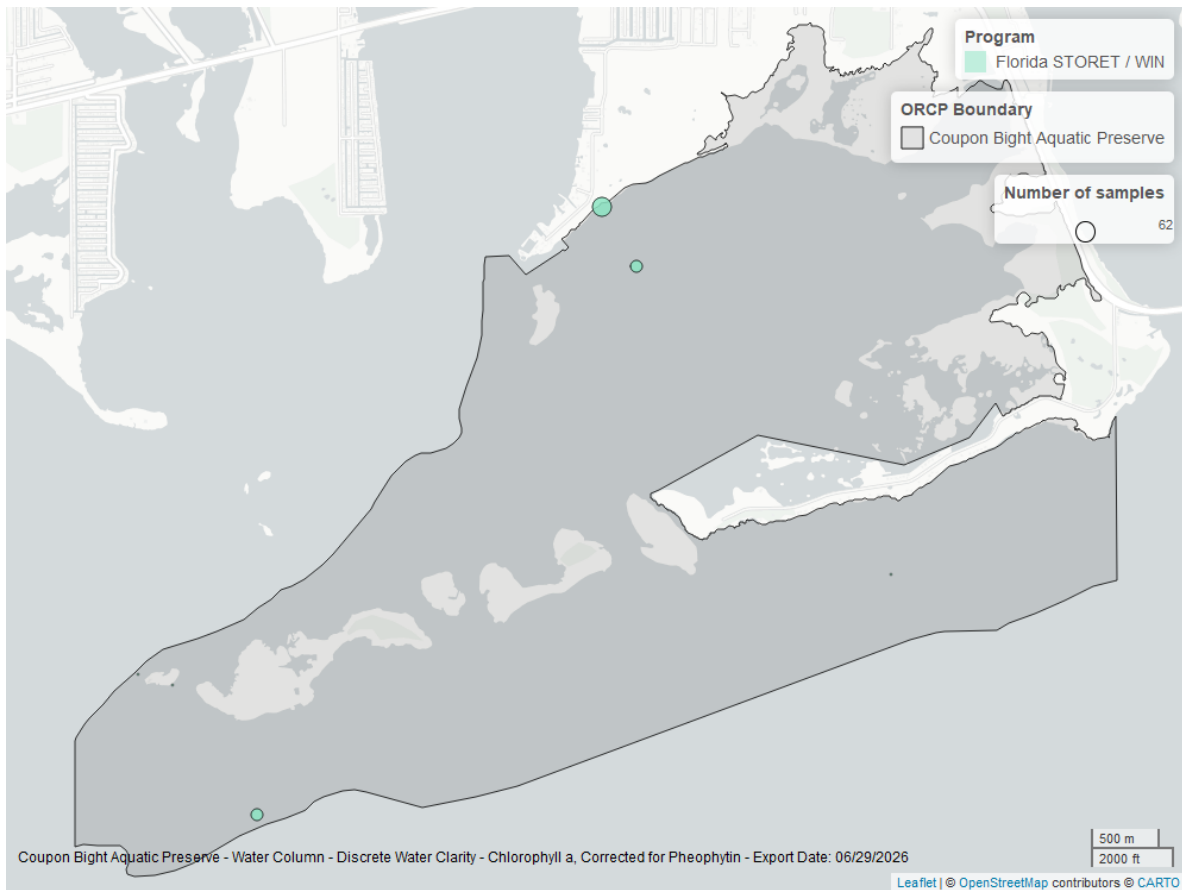


Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

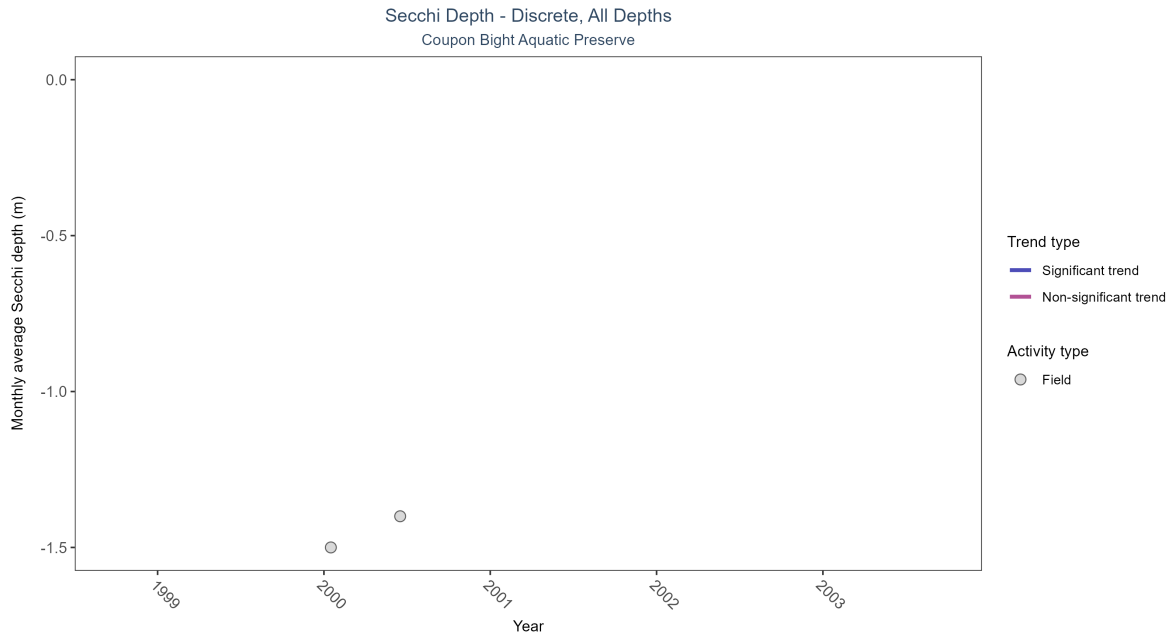


Figure 35: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 18: There was insufficient data to fit a model for Secchi depth.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Insufficient data	4	1	2000 - 2000	-1.45	-	-	-	-

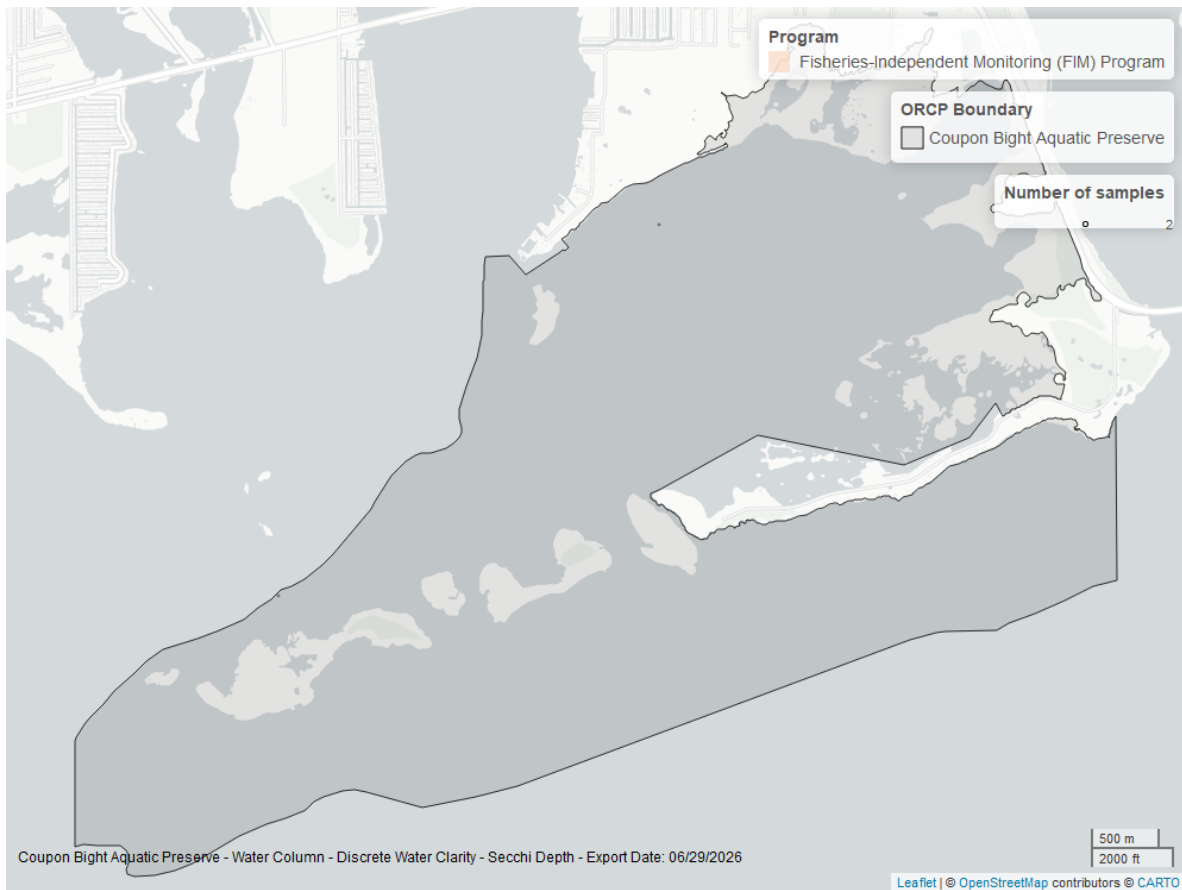


Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Coupon Bight Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.